

LatchMoE Architecture: Slot Virtualization for Graph-Compatible MoE Offloading

HOST (CPU + DRAM)

Pinned Expert Store

Complete expert pool in host DRAM



Dynamic Control Plane

- Routing decisions
- Cache policy
- Wave planning
- Slot assignment

Core Abstraction: Logical-to-Physical Mapping

Routing yields logical expert IDs; staging maps them to fixed physical slots

Logical ID	Physical Slot	Version
e	S	v
e	S	v
e	S	v

NPU (HBM + Compute)

Captured Graph Boundary (fixed addresses, re...

H2D Transfer

Captured Router

Router Op

Input: hidden states
Output: top-k IDs + weights

Example: [e , e , e]

Captured (fixed address)
Native routing logic
No approximation

Staging Boundary

(Uncaptured - Replay Boundary)

Cache Check & Wave Planning

For each routed expert ID:
• Hit? → Keep current slot
• Miss? → Assign new slot
Transfer from host
Increment version

Dynamic decisions
Bounded capacity

Captured Expert Computation

Fixed Slot Bank (replay-stable addresses)



stable refs

Fused MoE Kernel

Routed MLP + Combine
Reads from stable slot addresses

Key Insight: Logical ownership changes (), but graph-visible storage () remains fixed

Data flow

Control flow

Stable reference